

Bryce Richard

Machine Learning Engineer — Computer Vision, 3D Geometry, Perception Systems

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SUMMARY

Computer vision engineer building perception systems that run outdoors, on real hardware, at production scale — multi-view geometry, calibration, tracking and re-identification, and the services behind them. Mechanical engineering background, so the optics and the geometry come before the model.

EXPERIENCE

Blissway — Machine Learning Engineer

Denver, CO · June 2025 – Present

Founding member of the ML team at a Y Combinator-backed company replacing roadside tolling and safety hardware with vision. Own perception end to end: data collection and labeling strategy, model training, evaluation infrastructure, and the production services. Models run in the cloud and on NPUs mounted to poles on live interstates.

- Built the **vision-based vehicle triggering system** that replaces in-road induction loops: detection on a Hailo NPU feeding a multi-frame tracker that maintains vehicle identity across frames and decides when a vehicle has been fully captured. **0.10% unrecoverable miss rate at 4.9% false-positive inflation** over 4,791 ground-truth vehicles; 0.874 / 0.968 precision-recall on the device-faithful benchmark.
- Built **projective-geometry lane registration** that keeps each camera's stored lane model aligned as poles shake and cameras drift — vanishing-point and anchor optimization against segmented lane markings. Recovered lane coverage from **0% to 76%** on a drifted corridor; tracked camera motion to **1–3 px through 3° of rotation at ~15 ms/frame on CPU**, validated on real roadside footage.
- Built **axle counting as a multi-view fusion problem**: per-frame wheel detections from several cameras projected into a single virtual chassis per vehicle trip, with RANSAC outlier rejection and density clustering over the fused representation. **100% precision / 99.4% recall** on the billing-critical 3+ axle call, with ~1% of trips routed to human review.
- Wrote an **analytic camera-placement simulator** — pinhole projection plus multi-stage Nelder-Mead over aim, zoom and illumination — that answers roadside siting questions without a field trip, with its detection model calibrated against recorded field data to **reproduce measured recall within 5 pp** on replayed clips.
- Fine-tuned a **DINOv3 vehicle re-identification backbone** for cross-camera matching and vehicle fingerprinting, raising ReID mAP from **0.69 to 0.92** in a multi-task co-train while cutting backbone FLOPs **3.3×**.
- Trained a custom **multi-head license plate reader** (segmentation → rectification → OCR, state and design heads) that beats the commercial ALPR incumbent by **7.7 pp on plate text** (93.8% vs 86.1%) and **8.4 pp on state** (97.5% vs 89.1%).
- Re-architected the inference pipeline into **per-service microservices** with metered cost attribution, and swapped the segmentation tier to RF-DETR for **2.4× cheaper serving at equal accuracy** across ~3M events (~7M images) per day.

Carnegie Mellon University, Biorobotics Lab — Graduate Research Assistant

Pittsburgh, PA · August 2023 – May 2025

- Designed and implemented a **real-time sortation system for a robotic disassembly cell** in collaboration with a major OEM, training and fine-tuning YOLOv8, U-Net and Mask R-CNN for semantic and instance segmentation of components on a moving line.
- Integrated a **SCARA arm with vision-in-the-loop decision-making over ROS**, closing the loop from detection to pick on live hardware, and built a **Segment Anything-based annotation tool** that cut labeling time **50%**.
- Prototyped a tool to infer **forces and internal stresses in a smart device from high-speed X-ray video**, connecting image-space measurement to physical models.

Brigham Young University, FLOW Lab — Research Assistant

Provo, UT · August 2022 – May 2023

- Developed layout optimization tools integrating bathymetry data for offshore wind farms, **reducing projected LCOE by 10%**.

EDUCATION

Carnegie Mellon University — M.S. Mechanical Engineering (Research)

Pittsburgh, PA · May 2025 · QPA 3.9 / 4.0

Advanced Computer Vision · Intermediate Deep Learning · Trustworthy AI · Systems & Toolchains for AI Engineers

Brigham Young University — B.S. Mechanical Engineering, Minor in Computer Science

Provo, UT · April 2023 · GPA 3.9 / 4.0

SKILLS

Languages — Python (advanced), C++

Perception & 3D — projective geometry, camera calibration and registration, multi-view and temporal fusion, tracking, re-identification and metric learning, semantic and instance segmentation, OCR; PyTorch, OpenCV, ROS, DINOv3 / ViT, YOLO, RF-DETR, SegFormer, SAM

Systems & Deployment — edge inference on Hailo NPUs, ONNX, Docker, FastAPI, Modal, AWS, MongoDB, PostgreSQL, CI/CD, evaluation and monitoring infrastructure, SolidWorks